

The Decision Log

- **Component Refactoring:** Migrated the loose conversational script from `test_json.py` into a strictly defined, reusable functional module (`ai_brain.py`) capable of parsing dynamic user inputs.
- **Payload Serialization:** Upgraded the AI prompt structure to seamlessly bind live string variables extracted by the file system, enabling real-time generation of custom payload models.
- **Master Controller Implementation:** Engineered `main.py` as the system command center, successfully importing custom modules and strictly enforcing a 3-step pipeline sequence.
- **Data Typing Alignment:** Mapped the keys generated deterministically by the Gemini AI configuration directly to the exact string targets expected by the Playwright automation locators.
- **Graceful Termination Architecture:** Integrated the native Python `sys` library to handle validation failures defensively, ensuring the system shuts down cleanly instead of executing rogue browser commands during an upstream failure.
-